

Joseph Sheetz

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EDUCATION

Bachelor of Science in Physics - Expected May 2026

University of Northern Iowa - Cedar Falls, IA

- Enrolled in a 3+2 dual-degree program with Iowa State University.

Bachelor of Science in Electrical Engineering - Expected Fall 2027

Iowa State University - Ames, IA

- Part of the 3+2 dual-degree program with the University of Northern Iowa.
- Both degrees awarded upon completion of studies at Iowa State University.

Associate of Arts Degree - Graduated 2022

Marshalltown Community College - Marshalltown, IA

- Earned an Associate's degree while in high school.
- Excelled in humanities courses and received recognition for achievement in sociology.

PROFESSIONAL EXPERIENCE

John Deere – Technical Product Verification and Validation Engineer (PV&V) - Aug 2025 to Present

Ames, IA

- Designed and developed Python scripts to ingest and process test data for automated report generation.
- Led and authored test cases for verification and validation of autonomous tractor systems.
- Automated safety, regression, and quality testing for autonomous vehicle platforms.

John Deere – Product Development & Production Manufacturing Engineer - Nov 2023 to Aug 2025

Waterloo, IA

- Implemented vision systems for quality control across new and existing drivetrain production lines.
- Created and standardized an image storage process adopted by multiple engineering teams.
- Quoted, sourced, and deployed vision systems across factory operations.
- Received multiple On-The-Spot awards for initiative and technical performance.

Fareway Grocery Stores - Apr 2019 to Nov 2023

Cedar Falls, IA

- Advanced through three roles with increasing responsibility.
- Doubled hourly wage through performance-based promotions.

PROJECTS

Automated End-of-Line Inspection System

- Designed and implemented an automated inspection system for 6,000 lb rear axles using an autonomous cart, cobot arm, and 3D blue-light laser scanner.
- Achieved near 100% defect detection accuracy with a platform adaptable to any factory line.
- Led hardware research, process integration, and validation testing.
- Secured \$500K+ in funding and delivered a fully validated system now deployed in production.

Vehicle Automated Test Revamp

- Revamped automated vehicle testing for autonomous tractor development.
- Expanded testing scenarios through custom script development.
- Collected and interpreted sensor data to determine pass/fail outcomes.
- Integrated tests into the CI pipeline for validation on every software release.

Digital Timer

- Designed a digital timer circuit in Multisim using digital logic components.
- Implemented counter chips and logic gates driving six 8-bit hexadecimal displays.
- Added user input controls and a switch-triggered countdown mechanism.
- Integrated an LED indicator that flashes when the countdown reaches zero.

Data Ingestion and Processing Script

- Developed a Python script to automate data ingestion from Resim.AI using API integration.
- Implemented data cleaning, formatting, and Excel export for reporting.
- Scheduled automated execution for monthly report generation.
- Improved efficiency and consistency of performance tracking workflows.

TECHNICAL AND PROFESSIONAL SKILLS

Programming Languages: Python, C, MATLAB

Software: Microsoft Office Suite, Power BI, LTspice, Multisim, GitHub, GitLab

Engineering Experience: Automation systems, vision systems integration, data processing, test implementation and creation, hardware testing, report writing, hardware acquisition, specification development, supplier collaboration, and global engineering operations

LEADERSHIP AND ACTIVITIES

Physics Club Executive - University of Northern Iowa

- Served as Treasurer and Social Media Manager, organizing outreach and engagement initiatives.
- Promoted STEM activities and coordinated student-focused events.